AI-Based Password Strength Analyzer with User Authentication System

Abstract

This project focuses on developing a secure password strength analyzer using machine learning and rule-based techniques. It includes user authentication, OTP-based password reset, and an admin dashboard for monitoring users.

Introduction

Passwords are the first line of defense in cybersecurity. Weak passwords can lead to data breaches. This system helps users create strong passwords and provides secure authentication features.

Objectives

- Analyze password strength
- Implement login/signup system
- Provide OTP-based password reset
- Build admin dashboard

Methodology

The system uses TF-IDF vectorization and Logistic Regression for password strength prediction. It also applies rule-based validation and integrates email OTP verification.

System Architecture

User Interface (Streamlit) → Backend Logic → ML Model → Database (JSON)

Modules

1. Login System
2. Signup with OTP
3. Password Analyzer
4. Admin Dashboard
5. Email OTP System

Technologies Used

Python, Streamlit, Scikit-learn, Pandas, SMTP

Results

The system successfully classifies passwords and provides secure authentication with OTP verification.

Conclusion

This project enhances password security using AI and demonstrates practical cybersecurity implementation.

Future Scope

- Password encryption
- OTP expiry
- Deployment on cloud
- Advanced analytics